



7301 - A New Modeling Toolkit for JWST Ice Observations

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

JWST is unique in its ability to probe the composition and structure of cosmic ice, particularly the dust grain ice mantles found in star-forming regions. Nevertheless, the fraction of JWST programs making use of this critical capability is still small. One reason for the "glacial" pace of JWST-enabled ice science is perhaps the current difficulty of accurately predicting the molecular composition of ice in the interstellar medium: a critical component of planning new observations or analyzing data from existing studies. We propose to address this troubling barrier to progress through the development of new open source theoretical/computational tools which incorporate recent advances in modeling the chemical evolution of ice under astrophysical conditions. Specifically, we will take a tripartite approach to this problem, focusing on: how reactions occur within solids at low temperatures, the physics of UV photon and cosmic ray ice bombardment and, the underlying chemical reactions driven by energetic ice processing. These three objectives are closely interrelated and essential to consider together, since Objective 2 describes how reactive species are made in ice, Objective 1 describes how those species react, and Objective 3 describes which reactions occur. As a theory project, our work will yield well-documented, publicly available tools that unleash the power of JWST as an ice observatory, all without taking up any valuable observing time. The proposed modeling toolkit will not only represent the state of the art in the astrochemical modeling of ice in the ISM, but will also be of great interest to the study of icy bodies in our Solar System and exoplanetary systems.

OBSERVING DESCRIPTION